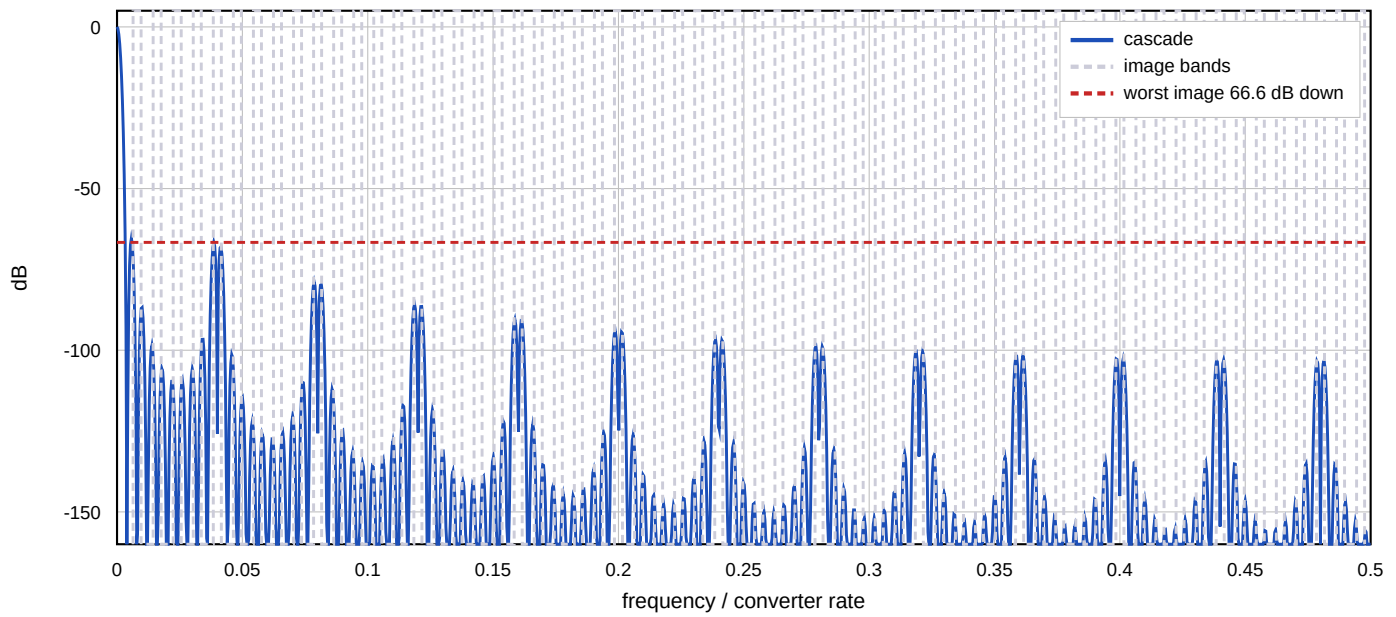


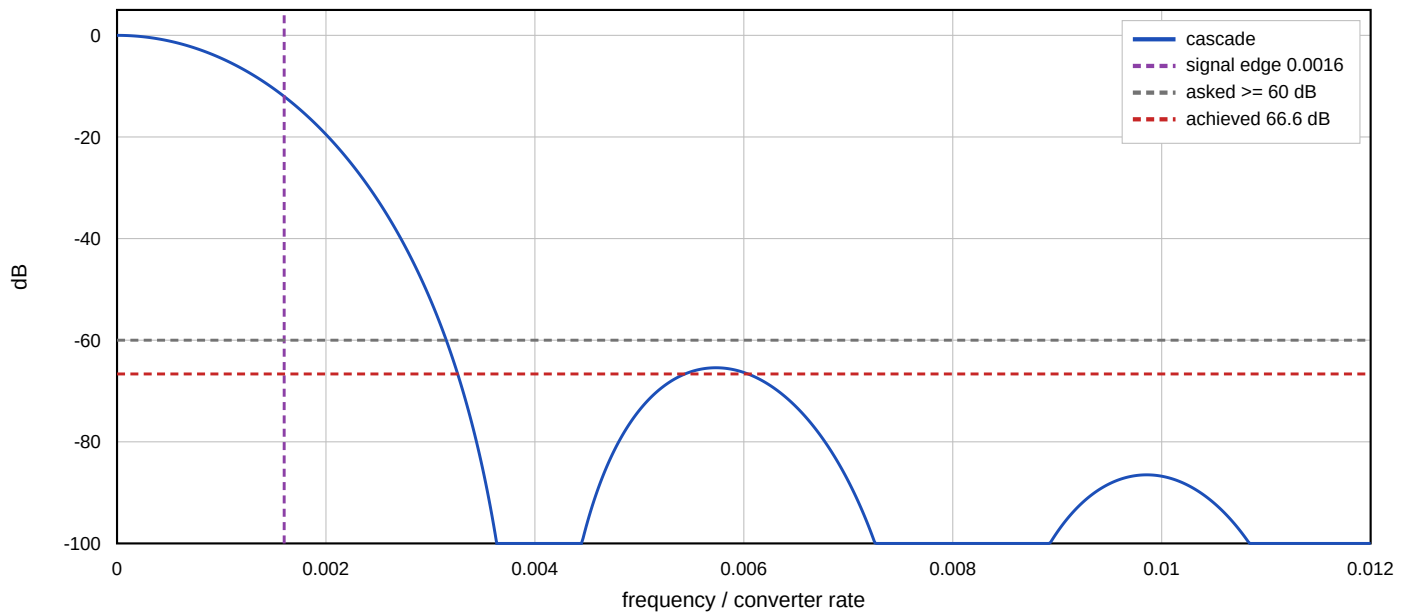
Interpolation Chain - Derived Design

1000.0000 kHz x 125 = 125.000 MHz, split [5, 25], 200.000 kHz signal

Composite magnitude across the converter band



The signal and its first image



Stages

stage 1: x5 N=5 M=2 combs at 1.0000 MHz, integrators at 5.000 MHz
in 16 bits, uniform accumulator 31 bits, tapered widths [17, 18, 19, 20, 21, 21, 24, 26, 28, 31]
465 register bits uniform, 320 tapered (lossless)

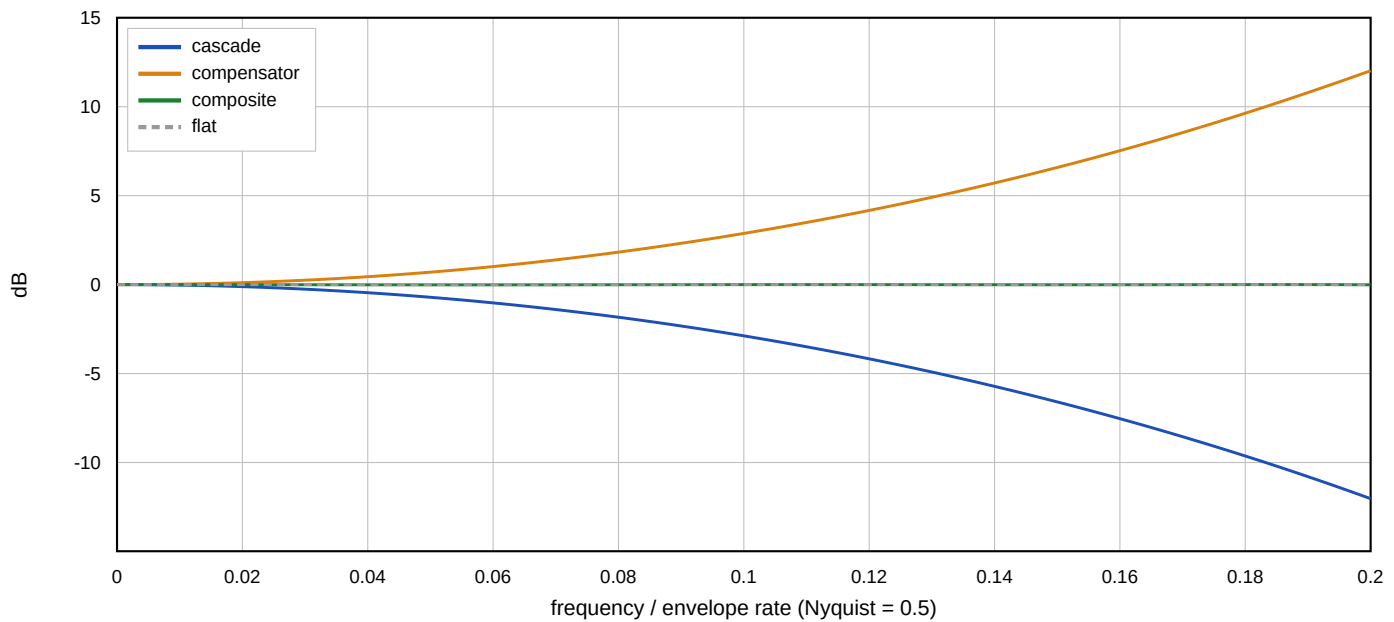
stage 2: x25 N=2 M=1 combs at 5.0000 MHz, integrators at 125.000 MHz
in 31 bits, uniform accumulator 36 bits, tapered widths [32, 33, 32, 36]
144 register bits uniform, 133 tapered (lossless)

The combs run at the stage's input rate and the integrators at its output rate, so an integrator bit costs R times what a comb bit costs. Depth belongs early and rate late. The tapered widths are lossless: each stage sized to its own growth bound holds its value exactly, so a tapered interpolator is bit-identical to a uniform one.

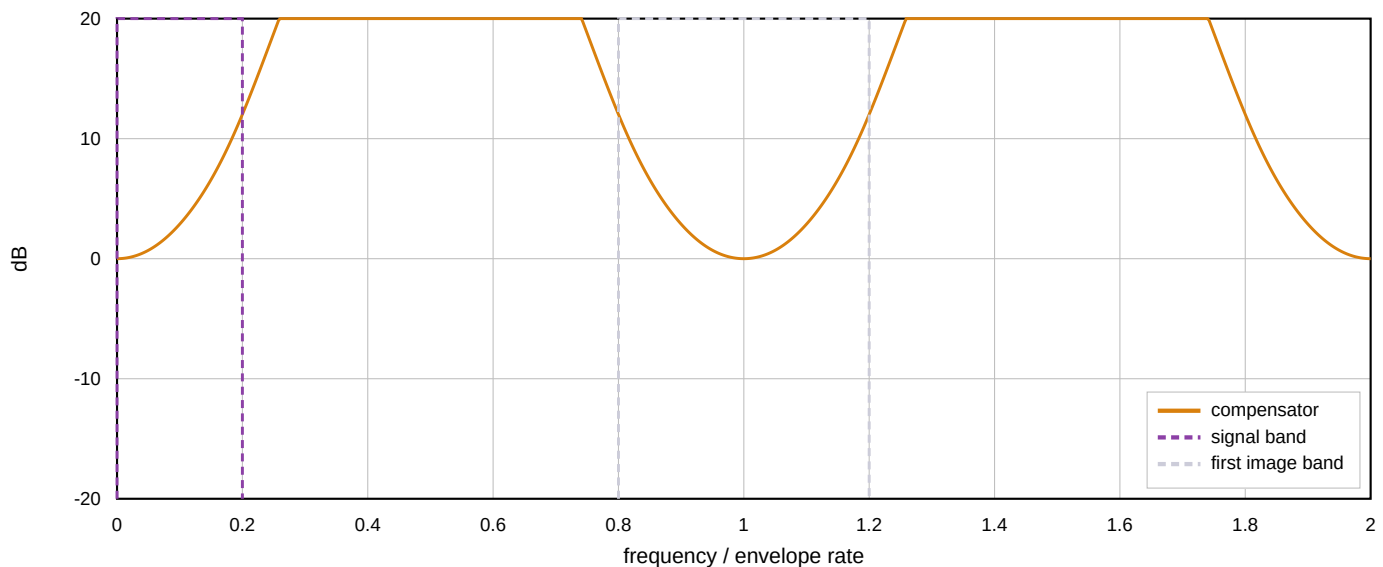
Compensation and Result

9 taps, 16-bit coefficients, at the 1.0000 MHz envelope rate

Droop, pre-compensator, and the two together - the signal band



The compensator repeats, so it lifts every image as much as the signal



Figures

images 66.6 dB down (asked ≥ 60.0)
worst image at 4.8297 MHz
passband ripple 0.0147 dB (asked ≤ 0.100)
uncompensated droop .. -11.986 dB at the band edge
converter floor 86.0 dB at 14 bits (the chain adds none)
register bits 609 uniform, 453 tapered
compensator DC gain .. 1.0000, shift 10
runner-up split [5, 25] N=[5, 2] M=[2, 2], 693 register bits
taps [304, -2290, 8288, -18223, 24866, -18223, 8288, -2290, 304]

More taps will not improve image rejection.

The compensator runs before the rate change, at the envelope rate, so its response is periodic and the image at $k+u$ sees exactly the gain the signal at u sees -- the plot above shows it. The image-to-signal ratio is the cascade's alone. This is the one place where transposing receive-side intuition is wrong: there, the compensator sits after the fold and its stopband is part of the alias budget.

The knobs are the CIC depth N , the interpolation factor R , and the signal bandwidth. A compensator at the converter rate would work, because it is no longer periodic in u , and costs a FIR running at the full clock.